EXP_FB984821 · PRE-REGISTERED DESIGN BRIEF · OPERATOR

GROWTH · CHECKOUT · SEALED 2026-06-05T18:43:44.845392Z · DURATION ~7.7 DAYS AT 7,735/DAY ACCRUAL

Buy now above the fold

Rendering the Buy Now call-to-action above the fold on the product detail page, relative to the current below-fold placement, increases conversion_rate among product-page visitors on the demo storefront over the exposure window, with the expected direction being a higher per-user probability of placing at least one order. The treatment is hypothesized to act by reducing the interaction distance between product consideration and purchase initiation; the magnitude of the effect is pre-registered in this brief at the feasibility-bounded relative MDE below and is not asserted as a point prediction.

> AT A GLANCE
>
> **Primary metric is *conversion_rate*; decision is binary against a positive lower bound on conversion lift AND no guardrail breach.**
>
> *Pre-registered before any analysis. Decision is binary against conversion_rate: a 7.0% relative relative lift over a 10.0% baseline requires 29,718 per arm at the standard a / power settings. Guardrails: revenue_per_user, page_load_time, cart_abandonment_rate.*

SAMPLE REQUIRED (PER ARM)

29,718

to detect a 7.0% relative relative effect at a = 0.05, power = 0.80

BASELINE

10.0%

DURATION

~7.7 days at 7,735/day accrual

TOTAL SAMPLE

59,436

FEASIBILITY

VIABLE

MECHANISM

Why we think this works

An above-fold Buy Now call-to-action reduces the steps between product consideration and purchase initiation; the treatment is expected to lift the per-user probability of placing at least one order.

Conversion baseline on the demo storefront sits near 10% over the trailing 28 days.

METRICS

conversion_rate is the verdict-driving metric.

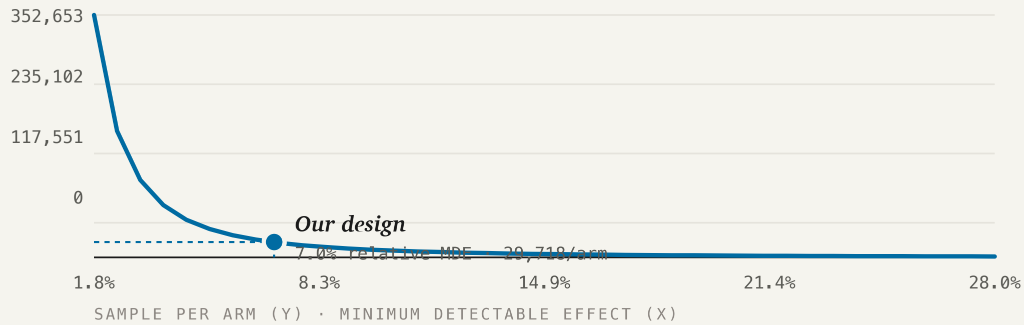
Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their CI crosses the pre-registered non-inferiority margin in the adverse direction.

PRIMARY	**conversion_rate** Proportion · baseline 10.0% · MDE 7.0% relative · direction higher is better
GUARDRAIL	**revenue_per_user** Direction higher is better. Decision blocks if 90% CI lies past -1.0% relative.
GUARDRAIL	**page_load_time** Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative.
GUARDRAIL	**cart_abandonment_rate** Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 7.0% relative relative effect against a 10.0% baseline, the design requires 29,718 per arm. Tighter detection thresholds climb the curve to the left.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the user_id level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.

UNIT	**user_id** One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed.
ARMS	**50 / 50** control · treatment
COHORT	**Users on the demo storefront whose first exposure to the experiment is their first `page_event` row with `event_name = 'product_view'` during the exposure window; assignment fires at that event (`assignment.first_exposure_at = MIN(page_event.event_at WHERE event_name = 'product_view')` per `user_id`), so "all assigned users" is by construction the set of product-page visitors during the window, segment = general.**

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Ship the treatment if and only if the two-sided 95% confidence interval on the relative lift in conversion_rate excludes zero on the positive side (lower bound strictly greater than zero) AND no guardrail breaches its pre-registered non-

inferiority margin. If the primary interval crosses zero, or if any guardrail's confidence interval crosses its non-inferiority margin in the adverse direction, the verdict is do-not-ship regardless of point estimate. The rule is intentionally tight: a treatment whose true effect is below the pre-registered MDE is expected to fail it.

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width > 2 × MDE
INCONCLUSIVE	Underpowered ($n <$ required) AND primary CI straddles zero
INVALID	SRM χ^2 $p < 0.0005$ and no override resolved — investigate assignment